

Learning Week: Day 1 → Day 6

A complete guide to understanding chains, agents, memory, orchestration, and building a real retail data pipeline



Day 1: Chains

Day 2: Templates

Day 3: Memory & Tools

Day 4: LangGraph

Day 5: Orchestrator

Day 6: The Project

Day 4: What is LangGraph?

Building Decision-Making Systems

Chain vs. Agent (vs. Graph)

A **chain** is linear — always the same steps in the same order.

An **agent** is intelligent — it can decide what to do next based on the situation.

A **graph** is how we build these decision systems. It's a visual way to represent: nodes (steps), edges (connections), conditional edges (decisions), and state (data flowing through).

Core LangGraph Concepts

Node: A single step or function in the graph

Edge: A connection showing what happens next

Conditional Edge: A decision point ("if result is X, go to node Y")

State: The data passed along and updated as it moves

Graph: The entire system of nodes, edges, and state

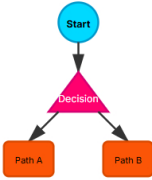
Chain vs. Graph vs. Agent

Chain (Linear)



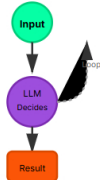
Always same order

Graph (Conditional)



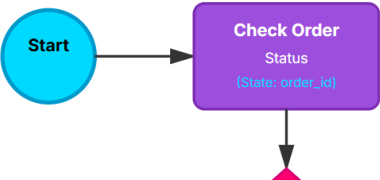
Can branch based on logic

Agent (Dynamic)



Loops until solution found

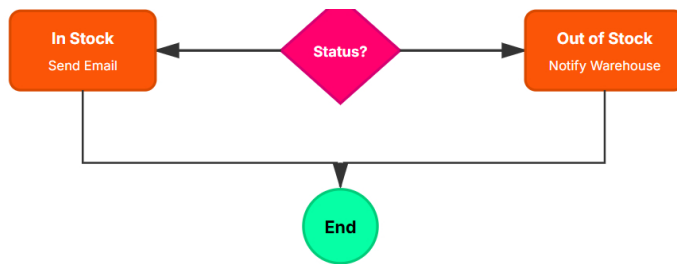
Example: Order Status System (LangGraph)



How it Works

- Start / End Node
- Process Node
- Decision Node
- Action Node

Edges = transitions
Conditional edges =
different paths based
on state / decision



Why LangGraph?

Chains are too simple — they can't handle "if this, then that" logic.

Agents are flexible but hard to visualize and debug.

LangGraph gives you a visual, structured way to build intelligent systems. You can see the flow, add branching logic, and know exactly what will happen.

State in LangGraph

State is the "data package" that flows through your graph. As it moves from node to node, each node can read the state and update it

EXAMPLE STATE:

```
{
  "order_id": "ORD-12345",
  "status": "processing",
  "items": ["Laptop", "Mouse"],
  "customer": "Ahmed",
  "action_taken": None
}
```

Key Takeaway: LangGraph lets you build systems that can make decisions and follow different paths. It's perfect for workflows where "sometimes we do this, sometimes we do that" — like our upcoming orchestrator agent project.

Free Resources to Dive Deeper

LangChain Docs

Official documentation with API reference and guides

LangChain Academy

Free official courses on LangChain fundamentals

LangGraph Docs

Guide to building graphs and agents

Multi-Agent Tutorial

Real example of orchestrator agents in action

🔗 Learning Week: LangChain + LangGraph Fundamentals

DevSynt AI Automation Internship · Mentor: Afnan Shoukat

Quiz coming next — make sure you understand these concepts!